

Exploring Semantic Expression Disparities in Intragenerational and Intergenerational Communication: A Novel Perspective on Socioemotional Selectivity Theory

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The Socioemotional Selectivity Theory (SST) posits that older and younger adults have different life goals due to differences in perceived remaining lifetime. Younger adults focus more on future-oriented knowledge exploration and forming new friendships, while older adults prioritize present-focused emotional regulation and maintaining close relationships. While previous research has found these age differences manifest in autobiographical textual expressions, their presence in verbal communication remains unexplored. We recruited 36 older adults and 36 younger adults to form 12 younger adults–younger adults dyads, 12 older adults–older adults dyads, and 12 younger adults–older adults dyads engaging in three conversational scenarios. Analysis of communication transcripts using Linguistic Inquiry and Word Count revealed that in intragenerational communication, older adults used more present-focused and family-related words, and fewer future-focused, negative emotion, and knowledge acquisition words compared to younger adults, aligning with SST predictions. However, in intergenerational communication, younger and older adults showed similar patterns in future-focused, cognitive, and family-related language use. This suggests that participants adjusted their goal orientations to accommodate each other, mitigating the age differences proposed by SST and partially supporting the Interpersonal Cognitive Consistency Model. These findings indicate that while SST explains semantic expressions in intragenerational communication, group dynamics toward verbal consistency play a more vital role in intergenerational communication.

Public Significance Statement

This study reveals that in intragenerational communication, older adults used more present-focused and family words, while using fewer future-focused, negative emotion, and cognitive words compared to younger adults—a pattern consistent with the Socioemotional Selectivity Theory. In intergenerational communication, research indicates an alignment between different age groups in their use of language concerning future-focused, cognitive, and family words. These findings provide valuable insights for developing nuanced cross-generational communication strategies that can potentially enhance intergenerational understanding and satisfaction.

Keywords: Socioemotional Selectivity Theory, communication, Linguistic Inquiry and Word Count, age difference, Interpersonal Cognitive Consistency Theory

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Aging is a global demographic trend. By 2050, the global population of individuals aged over 60 is projected to reach approximately 2 billion, nearly equaling the number of those under 15 (Harper, 2014). According to the World Health Organization (2022), the proportion of people aged 60 and over is growing faster than other age groups, with their population expected to increase substantially in the coming decades. As societies age, multigenerational communities become increasingly common, presenting both opportunities and challenges for interaction between different age groups (Zhong et al., 2020). Therefore, understanding the diverse needs and perspectives of various generations and fostering effective intergenerational communication are crucial for building cohesive future societies.

Age Differences in Life Goals Proposed in Socioemotional Selectivity Theory

The Socioemotional Selectivity Theory (SST) illuminates how time perspective shapes life goals across different age groups. As individuals age, they become increasingly aware of their limited remaining lifetime, which fundamentally alters their motivational priorities (Carstensen, 2006). This temporal awareness creates distinct goal-oriented behaviors between older and younger adults. Older adults, perceiving a shorter time horizon, prioritize emotional goals that emphasize immediate feelings and social connections. They focus on enhancing emotional satisfaction and sense of belonging, selecting experiences that provide meaningful interpersonal and emotional rewards. In contrast, younger adults, with a perception of expansive future time, prioritize knowledge-oriented goals. They are more likely to seek information, explore opportunities, and optimize long-term potential (Carstensen, 2021; Carstensen et al., 1999). These motivational differences manifest across multiple domains, including motivation, cognition, and emotional experiences. The theory suggests that goal selection is not merely age-dependent, but fundamentally driven by an individual's perceived time perspective.

van der Goot et al. (2019) delineated three primary age-related differences between emotional and knowledge goals. The first difference centers on *present* versus *future* orientation: Older adults prioritize present experiences to enhance life satisfaction, while younger adults are more inclined to optimize future prospects and engage in forward planning (Fung & Carstensen, 2003). The second difference involves *emotion regulation* versus *knowledge acquisition*: Older adults demonstrate a marked preference for regulating emotional states, seeking positive experiences, and avoiding negative emotional contexts more actively than younger adults (Carstensen et al., 2003). The *positivity effect* provides a mechanism for understanding the enhanced emotion regulation in older age. This phenomenon describes older adults' tendency to focus more on and better remember positive information relative to negative information compared to younger adults (Carstensen & Mikels, 2005; Mather & Carstensen, 2005). A meta-analysis by Reed et al. (2014) further validated this effect across diverse cognitive assessments, including autobiographical memory tests and visual attention to emotional stimuli. Conversely, younger adults maintain a stronger focus on knowledge acquisition, emphasizing learning opportunities and the potential for new knowledge expansion (Löckenhoff & Carstensen, 2004). The third distinction emerges in social relationship dynamics: Older adults invest more time in consolidating existing, emotionally fulfilling close

relationships, whereas younger adults tend to prioritize establishing new social connections with potential future benefits (Fredrickson & Carstensen, 1990).

Autobiographical Narratives and Communication Process Explained by SST

Age differences described by SST as found within autobiographical narratives may manifest in interpersonal communication among adults. Previous research has found that young and older adults' positive or negative autobiographical texts exhibit distinct semantic expression patterns corresponding to SST. Using the Linguistic Inquiry and Word Count (LIWC) tool, researchers have uncovered nuanced differences in emotional and relational language use. Robertson and Hopko (2013) found that younger adults employed more negative emotional words and friend-related words, while older adults predominantly used family-related words. Memory recall studies further illuminate these linguistic distinctions. When recounting their happiest and saddest memories, older adults demonstrated a remarkable ability to reframe negative experiences more positively compared to younger adults (Ikier & Duman, 2022). This pattern of emotional modulation was consistent across recent autobiographical memories (Schryer et al., 2012).

Robertson and Swickert (2018) provided additional insight by examining emotional expression during negative experience sharing, revealing that older adults used significantly fewer angry emotional words compared to their younger counterparts. A text analysis by Pennebaker and Stone (2003) synthesized these findings, demonstrating that older adults' narratives systematically differ from younger adults' in several key linguistic dimensions: Older adults used more positive words, fewer negative words, more present-focused words, fewer future-focused words, and fewer cognitive-related words.

Autobiographical memory sharing serves as a fundamental mechanism for exploring and maintaining intimate relationships (Beike et al., 2016). This narrative practice permeates diverse communication contexts, from recounting shared experiences with close companions to narrating events involving friends and establishing connections with new acquaintances (Selwood et al., 2020). The social vocabulary embedded in autobiographical texts extends beyond mere storytelling, demonstrating remarkable predictive power for individuals' social competencies. Research by Smorti et al. (2010) revealed that the linguistic characteristics of personal narratives can effectively forecast real-life social engagement, interpersonal abilities, and prosocial activities. Given the intricate relationship between personal narrative and social interaction, age-related differences observed in intrapersonal autobiographical memory are expected to manifest consistently within interpersonal communication patterns.

Potential Differences in Emotional Expression Between Intra- and Intergenerational Interactions

People's communication goals can be influenced by their communication partners. According to self-categorization theories, the group differences between younger and older adults proposed by SST may persist within an intragenerational communication context. Self-categorization theories posit that individuals naturally categorize others perceived as similar as in-group members, a fundamental social cognitive mechanism that reduces psychological

uncertainty. By identifying with similar others, people enhance their capacity to navigate complex challenges and manage organizational change more effectively (Hogg & Terry, 2000). Empirical research substantiates this theoretical framework, with studies revealing that older employees exhibit heightened workplace engagement when interacting with age-comparable colleagues (Avery et al., 2007). When individuals perceive communication partners from the same generation as in-group members, they typically anticipate shared goals and create a relatively safe communicative environment. In such contexts, people are inclined to express their authentic goals and needs more openly. However, SST emphasizes that the genuine goals and needs of younger and older individuals fundamentally differ. Consequently, within intragenerational communication, the age-related goal distinctions highlighted by SST are likely to persist.

In intergenerational communication contexts, however, the age-related differences proposed by SST may potentially converge through dynamic interpersonal processes. When communication partners possess divergent goals and perspectives, they can mutually influence each other's objectives, ultimately fostering a collaborative state of agreement. The foundational research on group dynamics provides insight into this convergence. Sherif's (1935) seminal work demonstrated that group members rapidly adopt collective norms and attitudes, leading to rapid consensus formation. Building upon this understanding, the Mutual Enhancement Model (Wittenbaum et al., 1999) revealed that team members preferentially share mutually relevant information, which generates positive interpersonal evaluations and reinforces collaborative discourse. Mojzisch et al.'s (2014) Interpersonal Cognitive Consistency Model further illuminates this phenomenon, suggesting that during cooperative interactions, individuals demonstrate a pronounced preference for communication that satisfies mutual preferences. This mechanism creates a sophisticated pathway toward goal alignment and shared understanding across generational boundaries.

In addition, The Communication Accommodation Theory provides a sophisticated framework for understanding interpersonal communication dynamics, proposing that interlocutors strategically adapt their communication styles to align with their partners' characteristics and needs (Giles et al., 2023). These adaptive strategies encompass multiple dimensions, including modifying communication abilities, selecting contextually appropriate topics, and negotiating role-based interactions that reflect mutual respect and understanding (Harwood, 2000). Empirical research illuminates the practical applications of Communication Accommodation Theory across intergenerational contexts. Bates and Taylor (2016) demonstrated how grandparents and grandchildren employ adaptive communication strategies to maintain familial social identity and enhance relational quality. Complementing this perspective, the Communication Predicament Theory offers additional insights into age-related communication nuances. Ryan et al. (1995) revealed that older adults are particularly attuned to negative aging stereotypes, motivating them to strategically adjust their communication—such as modulating voice volume or employing exaggerated nonverbal cues—to achieve more satisfactory intergenerational interactions.

The Present Investigation

This study aimed to utilize semantic analysis to capture vocabulary related to SST, which posits age differences in goal pursuits

(Carstensen et al., 1999; van der Goot et al., 2019), and explore whether the linguistic expressions of older adults and younger adults are different in three aspects: (a) *present* versus *future*, (b) *emotion regulation* versus *knowledge acquisition*, (c) *close* versus *novel social relationships*.

Regarding the first aspect, it is hypothesized that older adults will use more present-focused words and fewer future-focused words than younger adults (Hypothesis 1-a). Regarding the second aspect, older adults are expected to express fewer negative emotion words, more positive words, and fewer words related to cognitive processes and insight than younger adults (Hypothesis 1-b). Finally, regarding the third aspect, it is anticipated that older adults will use more words related to family and fewer words related to friends than younger adults (Hypothesis 1-c). Additionally, the study aimed to investigate whether the context of intergenerational or intragenerational communication affects the expression patterns predicted by SST, with the expectation that differences observed in intragenerational communication will diminish in intergenerational communication (Hypothesis 2).

Method

Participants

Recruitment was conducted through a web posting in the summer of 2022. Seventy-two participants were invited to join the study and were eligible to receive a reward of 360 New Taiwan dollars. All participants included 36 younger adults (YA; $M_{\text{age}} = 22.17$; age ranged from 18 to 37 years old; 18 women and 18 men) and 36 older adults (OA; $M_{\text{age}} = 68.75$; age ranged from 57 to 78 years old; 18 women and 18 men). All participants were Taiwanese and fluent in Mandarin Chinese. Eighty-six percent of the participants held a college or university degree or higher educational qualification. The study utilized the Montreal Cognitive Assessment (MoCA) to assess the basic cognitive level of older and younger adults, and all participants had a MoCA score greater than 26, which means that they can perform basic communication skills (Nasreddine et al., 2005). Building on Robertson and Hopko's (2013) study on the differences in family vocabulary expression between older and younger adults, we used the GLIMPES program to calculate the required sample size (Kreidler et al., 2013). The results showed that, in order to achieve a statistical power of 0.8 with a Type I error rate of 0.05, 72 participants in this study were sufficient to detect the interaction effects of dyad type and age.

Procedure and Design

Thirty-six younger adults (YA) and 36 older adults (OA; 36 females; 36 males) were recruited in this experiment, which formed 12 YA–YA dyads, 12 OA–OA dyads, and 12 YA–OA dyads. Among them, older adults were all over 55 years of age ($M = 68$, $SD = 4.9$), while younger adults were all under 40 years of age ($M = 22$, $SD = 3.3$). Each participant was randomly assigned to a dyad, and gender was counterbalanced both between age groups and within each dyad (12 females and 12 males within each group). In addition, the mean age for older adults (68.71) and young adults (22.42) in the intergenerational dyads is roughly equivalent with respect to the mean age of the older adults (68.83) and younger

adults (21.67) in the intragenerational dyads. For more details about the assignment, please see Supplemental Material 4.

After passing the MoCA screening and filling out a questionnaire on a controversial issue, participants went through a 90-min dialogue session. In Session A, they read a story about an athletic student and discussed it freely based on the story within a total of 15 min. In Session B, during the first 10 min, two interlocutors would take turns sharing their positive emotional stories. In the following 10 min, the two interlocutors would take turns sharing their negative emotional stories. That is, both participants were required to share one negative and one positive story within 20 min. In Session C, based on the answers to the controversial issue questionnaire filled by the participants, the experimenter assigned a controversial issue in which the two participants had different opinions based on a pretest questionnaire and invited them to have a free discussion based on this issue within a total of 30 min. For each session, after the experimenter had read the instructions, s/he would inquire whether the participants had any questions or if there was anything unclear about the tasks for the session. Once it was confirmed that both participants fully understood the experimental content, the session began.

To capture the complete picture of communication styles, Session A was designed to simulate the daily communication process of two people discussing other people's stories; Session B was designed to simulate how two people responded to each other's emotional experience; and Session C was designed to simulate the daily life process of two people arguing over an issue. Sessions A–C are conducted sequentially to allow the discussion topics to progress from superficial to deep. Hence, the time allotted for sharing between the interlocutors gradually increases from Sessions A (15 min) to B (20 min) and then to C (30 min; Figure 1, see Supplemental Material 1 for details of instructions).

Data Analysis

All three sessions were recorded and videotaped; the videos were translated into verbatim transcripts by AI intelligent software (Yating translator and Google translator) and proofread by four proficient native Mandarin Chinese speakers. Then, the transformed Chinese transcripts were segmented again using a word segmentation system (Ma & Chen, 2003). Furthermore, the Chinese version of LIWC (Version 2015) was used for the word count in the article (Lin et al., 2020). LIWC builds a dictionary containing 90 categories of vocabulary and further calculates the percentage of different categories of words to infer an individual's inner thoughts or mental processes (Pennebaker et al., 2015).

This study analyzed verbatim transcripts in two ways. The first way was to analyze the entire verbatim transcript with LIWC software to see the expression of vocabulary related to SST across three communication topics. The second method was to cut each person's verbatim into three topics and then analyze each person's performance in each topic with the LIWC software to see the percentage of word expression related to SST. After analyzing the word ratios of the verbatim scripts, all data were analyzed using R (Version 4.1.2) and RStudio (Version 2023.06.2). Additionally, this study employed the “Lmer” package in R (Version 1.1.29) to observe the effect of age on semantic expression using multilevel regression.¹

Transparency and Openness

Our sampling plan, all data exclusions, all manipulations, and all measures are described in the study, and we adhered to the *Journal of Psychology and Aging* checklist. Data were analyzed using R (Version 4.1.2) and RStudio (Version 2023.06.2). The package “Lmer” (Version 1.1.29) was used for analysis. This study's design and its analysis were not preregistered. All material and procedure were approved by the research ethics committee at National Taiwan University (NTU-REC-202007HS058); the project title in research ethics committee is “Age differences in communication patterns: the potential use of new technologies to help older adults.” In addition, all instructions used for the present study are listed in Supplemental Material 1 and all deidentified raw data and code were stored in the Open Science Framework platform at https://osf.io/uyg8e/?view_only=a4b459a182e24f3ba5753edec7343224 (Zeng & Yeh, 2024).

Results

This study analyzed the verbatim transcripts of 72 adults using LIWC's dictionary, achieving an average vocabulary capture rate of 88.63%. We focused on LIWC categories relevant to SST (Table 1),² including sense of time (present-focused, future-focused), emotional regulation (positive emotion, anxiety, anger, sadness), knowledge acquisition (cognitive processes, insight), and close relationships (family, friend).

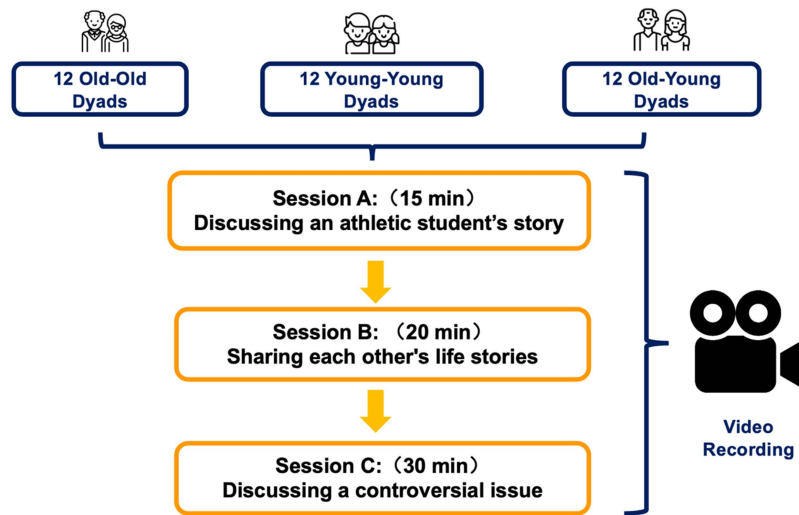
Differences in Semantic Expression Between Older Adults and Younger Adults

The experiment aimed to examine whether there are age differences in semantic expressions related to various aspects of SST. The results (Table 2) indicate that in terms of time horizon, we observed that older adults tended to use more present-focused words (estimate = -0.217 , $SE = 0.052$, $t = -4.164$, $p = .001$). In comparison, younger adults used more future-focused words (estimate = 1.407 , $SE = 0.202$, $t = 6.967$, $p < .001$). Additionally, regarding the expression of emotions, younger adults significantly expressed more anxiety words (estimate = 0.093 , $SE = 0.033$, $t = 2.802$, $p = .007$) and angry words (estimate = 0.107 , $SE = 0.041$, $t = 2.597$, $p = .012$) compared to older adults; however, there was no age difference in sad words expression (estimate = 0.049 , $SE = 0.038$, $t = 1.302$,

¹ According to Kenny et al. (2006), in the basics of dyadic data analysis, even two individuals meeting for the first time in a lab setting share a common experience, thus violating the basic assumption of traditional analysis of variance: Independence (data from one person are unrelated to data from another person). Multilevel regression is an appropriate analysis method to address this issue. This involves controlling for dyadic effects at Level 2 as a random effect and analyzing individual effects at Level 1 as a fixed effect. All results in this study were analyzed using this method.

² The LIWC software has undergone several changes across different versions (LIWC2001, LIWC2007, LIWC2015, LIWC2022). In the LIWC2001 version, the category of Positive Emotion included two subcategories: Optimism and Positive Feelings. However, by the time of LIWC2007's development, it was determined that these two subdimensions had consistently low base rates and were rarely used. Consequently, only the broad “Positive Emotion” category was retained without further subdivisions into more subcategories. In contrast, the subcategories of Negative Emotion (Anger, Anxiety, Sadness) performed well in reliability and validity tests; therefore, they have been retained in subsequent versions of LIWC. Since this study uses the Chinese version of LIWC 2015, the dictionary includes only the “Positive Emotion” category and does not further subdivide into more subcategories.

Figure 1
Three Sessions of the Experiment Procedure



Note. See the online article for the color version of this figure.

$p = .197$). In addition, the expression of positive emotion words showed no significant difference across different ages (estimate = -0.156 , $SE = 0.158$, $t = -0.989$, $p = .326$). Moreover, in the domain of knowledge exploration, younger adults expressed more cognitive process words (estimate = 3.03 , $SE = 0.526$, $t = 5.755$, $p < .001$) and more insight words (estimate = 0.722 , $SE = 0.184$, $t = 3.926$, $p = .001$). Finally, regarding relationship concerns, there were no differences observed between younger and older adults in the expression of family words (estimate = -0.174 ,

$SE = 0.123$, $t = -1.411$, $p = .164$) or friend words (estimate = 0.067 , $SE = 0.06$, $t = 1.132$, $p = .262$).

Age Differences in Semantic Expression Influenced by Communication Context

We further examined whether the differences mentioned in SST would vary depending on whether the communication occurred within the same generation or across generations. Results (Figure 2)

Table 1
Descriptive Statistics of LIWC Variables Related to SST

LIWC variable	<i>M</i> (%)	<i>SD</i> (%)	Range (%)	Example word in English and Chinese version
Time perception				
Present-focused	0.72	0.26	0.19–1.42	Currently, now (目前, 現今)
Future-focused	2.39	1.11	0.62–5.36	Soon, future (即將, 將來)
Emotion expression				
Positive emotion	3.18	0.69	1.86–5.84	Confidence, satisfaction, blessing (信心, 滿足, 祝福)
Negative emotion				
Anxiety	0.28	0.15	0.04–0.82	Restless, struggling, tense (不安, 掙扎, 緊繃)
Anger	0.33	0.19	0–0.90	Hate, complain, destroy (可惡, 抱怨, 破壞)
Sad	0.31	0.16	0–0.98	Heartache, frustration, powerlessness (心痛, 沮喪, 無力)
Knowledge acquisition				
Cognitive process	19.92	5.07	14.56–26.49	Understand, choose, doubt (理解, 選擇, 質疑)
Insight	4.56	1.23	2.27–6.40	Comprehend, realize, experience (了解, 恍然大悟, 體會)
Close relationships				
Family	1.03	0.72	0–3.90	Grandma, brother, granddaughter (祖母, 兄弟, 孫女)
Friend	0.41	0.28	0–1.52	Buddy, friendship (兄弟, 朋友)

Note. The values presented in the table are all percentages, calculated by dividing the number of words in the category by the total number of expressing words. The C-LIWC retrieves the verbatim transcripts of participants in Chinese. For ease of understanding, the corresponding English terms for the Chinese words are included in the "Example word" column. Based on considerations of base rates and usage frequency from the development of LIWC, only negative emotions are subdivided into three subcategories (Anger, Anxiety, and Sadness), while positive emotions are not differentiated into distinct subcategories (Pennebaker et al., 2015). $N = 72$. LIWC = Linguistic Inquiry and Word Count; SST = Socioemotional Selectivity Theory; C-LIWC = Chinese Linguistic Inquiry and Word Count.

Table 2*Differences in the Percentage of LIWC Words Between Younger and Older Adults*

LIWC variable	Younger adult	Older adult	<i>t</i>	<i>p</i>
	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)		
Time perception				
Present-focused	0.591 (0.200)	0.852 (0.270)	-4.164	.001**
Future-focused	3.090 (0.936)	0.591 (0.254)	6.967	<.001***
Emotion expression				
Positive emotion	3.130 (0.618)	3.230 (0.757)	-0.989	.326
Anxiety	0.324 (0.169)	0.229 (0.107)	2.802	.007**
Anger	0.386 (0.175)	0.278 (0.168)	2.597	.012*
Sad	0.334 (0.173)	0.280 (0.148)	1.302	.197
Knowledge acquisition				
Cognitive process	21.6 (2.04)	18.3 (2.35)	5.755	<.001***
Insight	4.92 (0.907)	4.20 (0.596)	3.926	.001**
Relationships				
Family	0.738 (0.586)	1.32 (0.723)	-1.411	.164
Friend	0.436 (0.250)	0.388 (0.303)	1.132	.262

Note. The values of mean and standard deviation in the table are all percentages calculated by dividing the number of words in the category by the total number of expressing words. This table shows the main effects of different ages on the prediction of LIWC vocabulary use, without considering the interaction of different communication groups and communication topics. The *t* test in the table is used to test the results obtained from the multilevel regression analysis, where the intercept is set as a random effect to control for variation between each dyad. *N* = 72. LIWC = Linguistic Inquiry and Word Count.

p* < .05. *p* < .01. ****p* < .001.

revealed an interaction between communication context and age in predicting future-focused (estimate = 1.705, *SE* = 0.381, *t* = 4.476, *p* < .001) words. Post hoc comparisons revealed that within the intragenerational communication context, older adults expressed fewer future-focused words (estimate = 1.975, *SE* = 0.206, *t* = 9.581, *p* < .001) than younger adults. However, within an intergenerational communication context, there was no significant difference between older and younger adults in the expression of future-focused (estimate = 0.271, *SE* = 0.389, *t* = 0.696, *p* = .494) words. Besides, there was a significant interaction between communication context, and age in predicting the expression of cognitive process (estimate = 2.064, *SE* = 1.034, *t* = 1.997, *p* = .049) and family (estimate = -0.909, *SE* = 0.276, *t* = -3.291, *p* = .001) words. Post hoc comparisons showed that within the intragenerational communication context, older adults expressed less cognitive process words (estimate = 3.953, *SE* = 0.602, *t* = 6.571, *p* < .001) and more family words than younger adults (estimate = -0.886, *SE* = 0.162, *t* = -5.459, *p* < .001). However, within the intergenerational communication context, there was no significant difference between older adults and younger adults in the expression of cognitive process words (estimate = 1.889, *SE* = 1.084, *t* = 1.743, *p* = .109) and family words (estimate = 0.023, *SE* = 0.158, *t* = 0.148, *p* = .885). The interaction effect between age and communication context was not found in any of the expressions of the other words. This result showed that communication partners adjusted their own goal orientations to accommodate each other and reached consistent word expressions related to future-focused, cognitive process, and family words.

We also explored whether the younger generation was making compromises for the older generation or vice versa during the dyad communication. Regarding the expression of future-focused words, we found that younger adults in intergenerational communication contexts used fewer future-focused words compared to those in intragenerational communication contexts (estimate = -0.710, *SE* = 0.310, *t* = -2.294, *p* = .033). Conversely, older adults in intergenerational communication used more future-focused words than those in intragenerational communication contexts (estimate = 0.994, *SE* = 0.293, *t* = 3.388, *p* = .002). However, concerning the expression of the cognitive process words, we only observed that older adults in intergenerational communication used more cognitive process words compared to those in intragenerational communication contexts (estimate = -1.752, *SE* = 0.879, *t* = -1.994, *p* = .064, marginal effect), while there was no difference between younger adults in intergenerational and intragenerational communication in terms of expressing cognitive process words (estimate = 0.311, *SE* = 0.818, *t* = 0.381, *p* = .707). Regarding the expression of family words, we found that older adults in intergenerational communication contexts tended to use fewer family words compared to those in intragenerational communication contexts (estimate = 0.55, *SE* = 0.285, *t* = 1.929, *p* = .067, marginal effect), while there was no difference between younger adults in intergenerational and intragenerational communication contexts in terms of expressing family words (estimate = -0.36, *SE* = 0.264, *t* = -1.365, *p* = .187).

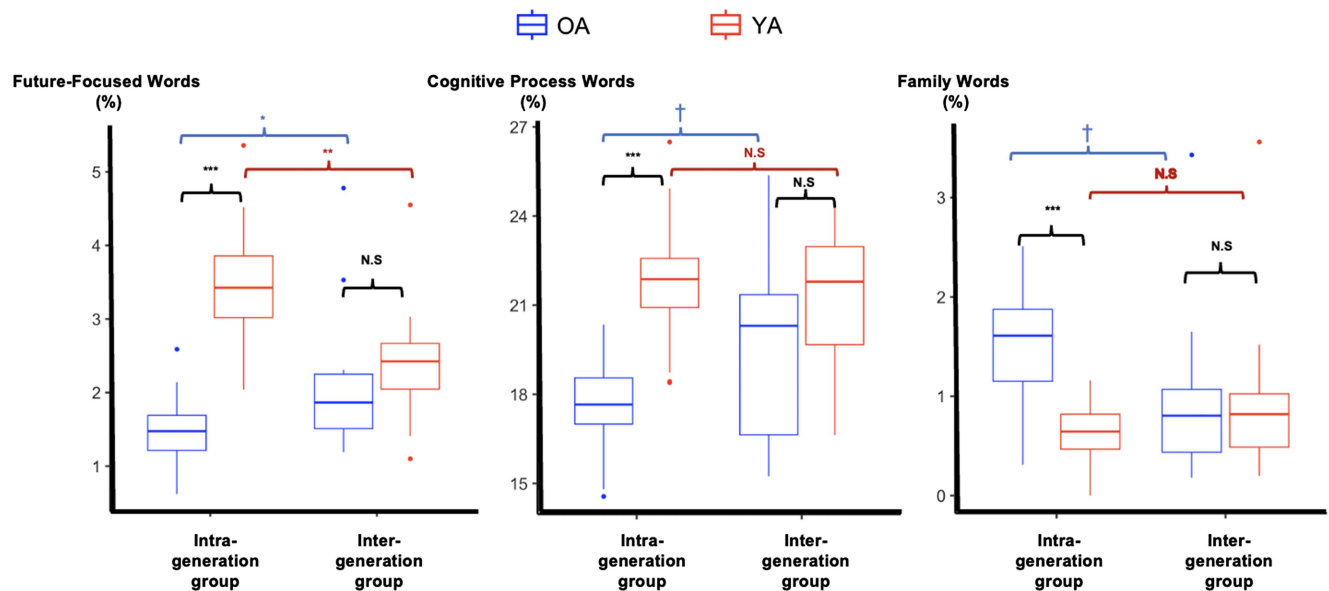
Examining Robustness by Controlling for the Influence of Communication Topics

We aimed to eliminate the influence of three communication topics and examine the stability of the aforementioned results. Thus, we statistically controlled for the influence of communication topics (Sessions A–C) and reexamined the effect of age on semantic expressions related to SST. The results found that differences between older and younger adults persisted in the expression of anxiety words, knowledge exploration (cognitive process words and insight words), and time perception (present-focused words and future-focused words), consistent with earlier described differences (for details, see Supplemental Material 2). Additionally, after controlling for the influence of communication topics, the interaction between communication context and age in predicting the use of future-focused words, cognitive process words, and family words remained significant, consistent with the results described above. Additionally, for other vocabularies that showed no significant main effects or interactions before controlling for the influence of communication topics, no significant trends were observed after controlling these topics (for details, see Supplemental Material 3).

Interestingly, after controlling for the influence of communication topics, only the age difference in anger-related vocabulary changed from significant (estimate = 0.107, *SE* = 0.041, *t* = 2.597, *p* = .012) to nonsignificant (estimate = 0.075, *SE* = 0.048, *t* = 1.580, *p* = .117). We divided it into three sessions for differential analysis to further verify the differences in expressing anger between older and younger adults on various topics. The results revealed that there was no significant age difference in expressing anger during Session A (discussing athlete-related stories; estimate = 0.007, *SE* = 0.102, *t* = 0.067, *p* = .947) or Session B (sharing emotional experiences;

Figure 2

Box Plots Representing the Difference in Verbal Expression Between Younger and Older Adults in Intrageneration and Intergeneration Groups



Note. These graphs illustrate the interaction between age and communication groups in predicting future-focused, cognitive process, and family words. The black line's bracket indicates whether the differences between older adults and young adults are significant within the intergenerational and intragenerational communication groups. The blue and red lines respectively indicate whether the differences within different communication groups are significant for young adults and older adults. The y-axis represents the proportion of Linguistic Inquiry and Word Count (LIWC) category vocabulary expression relative to the total expression count, and the x-axis represents intergenerational communication versus intragenerational communication. The blue box represents the effect for older adults, while the red box represents the effect for younger adults. $N = 72$. OA = older adults; YA = younger adults. See the online article for the color version of this figure.

N.S. $p > .1$. † $p < .01$. * $p < .05$. ** $p < .01$. *** $p < .001$.

estimate = 0.085, $SE = 0.06$, $t = 1.416$, $p = .161$). However, it was only during Session C (discussions on controversial topics) that younger adults tended to express more anger words than older adults (estimate = 0.181, $SE = 0.058$, $t = 3.133$, $p = .003$).

Discussion

This study analyzes verbatim transcripts of both intragenerational and intergenerational communication to examine age-related differences in vocabulary use as predicted by SST. Additionally, it investigates how communication context (inter- vs. intragenerational) influences these expression patterns. The results of this study support Hypothesis 1-a. Compared to younger adults, older adults expressed more present-focused words and fewer future-focused words. These findings align with SST, which posits that as individuals age, they increasingly prioritize present emotional experiences and derive meaning from immediate life contexts (Fung & Carstensen, 2003). These results complement previous analyses of autobiographical memory texts, which revealed that older adults tend to use more present-focused words when reminiscing, in contrast to younger adults' more future-focused narrative approach (Robertson & Hopko, 2013).

The results of this study partially support Hypothesis 1-b. Without controlling for the influence of communication topics, the results show that older adults expressed fewer words of negative

emotion (including anxiety and anger) and fewer words related to seeking knowledge (including cognitive processes and insight) than younger adults. The results regarding decreased negative emotion expression and knowledge seeking with age are consistent with SST (Carstensen, 2006; Carstensen et al., 1999). However, the absence of age-related differences in sadness expression aligns with Discrete Emotions Theory, which suggests that emotional expression varies across different emotion types (Kunzmann et al., 2014). While anger involves intense psychological arousal and requires substantial cognitive resources—more typically seen in younger individuals—sadness emerges from evaluations of irreversible loss, a common experience for older adults. Consequently, Discrete Emotions Theory predicts age-related decreases specifically in anger expression, but not in sadness expression.

Our findings revealed no significant differences in positive emotion word expression between age groups. The existing literature presents mixed perspectives on age-related positive emotional expression. While some studies suggest older adults express more positive emotions (Ikier & Duman, 2022; Pennebaker & Stone, 2003; Schryer et al., 2012), others find no age-associated variations (Labouvie-Vief et al., 2003; Pedder et al., 2016). Recent research proposes that the "positivity effect" described in SST operates primarily through attentional mechanisms rather than emotional experience. Specifically, although older adults demonstrate faster attentional responses to positive stimuli, this does not translate

to increased positive emotion reporting (Zsoldos & Hot, 2024). The observed lack of age differences in positive emotion word expression may be explained by the positivity effect's limited scope—functioning at an implicit attentional control level without significantly impacting explicit emotional communication.

The study explored the impact of communication topics on SST, revealing age-related differences in anger expression occurred only during controversial topic discussions. This finding aligns with previous research indicating that anger emerges most prominently during disagreements where parties' interests conflict (Van Kleef & Côté, 2007). Compared to personal emotional stories and others' narratives, contentious discussions are more likely to provoke anger, explaining why younger generations expressed significantly more anger words than older adults specifically in the controversial topic session.

This study's results do not support Hypothesis 1-c, as no age-related differences were observed in the use of family- or friend-related words. Our findings diverged from previous conclusions about age-based differences in autobiographical text expression (Robertson & Hopko, 2013). One possible explanation is that both participants engaged in the experiment were strangers meeting for the first time, and a prior meta-analysis study reveals that interactions between strangers often involve fear, potentially constraining discussions about intimate relationships (Sandstrom & Boothby, 2021). However, Sandstrom and Boothby (2021) demonstrated that initial interaction anxiety typically diminishes with successful communication, potentially allowing for more nuanced interactions over time. Therefore, it is possible that age differences in the expression of family- or friend-related words may require longer interaction times to become apparent.

Furthermore, the results of this study partially support Hypothesis 2, indicating that only in intragenerational communication do older adults appear to express more family, fewer cognitive process words compared to younger adults. However, there was no difference between older adults and younger adults regarding family and cognitive process words expression in intergenerational communication context. Peer groups tend to form more easily with in-group favoritism (McNamara et al., 2016). In intragenerational communication, shared social identity allows individuals to more comfortably maintain their original communication styles related to SST. These results support the Interpersonal Cognitive Consistency Model, Communication Accommodation Theory, and Communication Predicament Theory, wherein adults follow the principle of consistency in intragenerational communication contexts, adjusting their expressions to align with the other party's preferences to achieve a unified perspective (Giles et al., 2023; Mojzisch et al., 2014; Ryan et al., 1995).

The post hoc comparisons indicate that younger adults reduced their expression of future-focused words for the sake of older adults. Similarly, older adults also increased their expression of future-focused words for the sake of younger adults. These results support the Mutual Enhancement Model, which posits that both parties adapt to one another, fostering consistency and alignment throughout the communication process. In addition, the post hoc comparisons also reveal that older adults increased the frequency of expressing cognitive process words and decreased the frequency of expressing family words to accommodate the tendencies of younger adults (albeit with marginal effect), while younger adults did not adjust for older

adults. Research suggests that in intergenerational communication, younger individuals frequently harbor negative stereotypes about older adults (Harwood & Williams, 1998). However, when older adults make communicative accommodations, younger individuals tend to feel more supported, which can enhance the overall quality of interaction. As a result, older adults often adapt their communication style as a strategy to facilitate more effective dialogue and mitigate these preexisting stereotypes.

Limitations and Contributions

This study examined communication between strangers to minimize confounding variables from prior interactions. While everyday communication typically involves familiar relationships, research shows complex interpersonal dynamics, with adults exhibiting altruistic behaviors toward family and friends from childhood (Lu & Chang, 2018). However, potential intense family conflicts could significantly influence interaction patterns. Whether people continue to demonstrate the age differences depicted by SST in communication dialogues with more intimate relationships remains a topic worthy of further exploration in subsequent studies.

Relationship duration serves as a key indicator of relationship satisfaction (Gustavson et al., 2016). While our study's design progressed discussions from superficial to deeper topics, we acknowledge a limitation in definitively attributing the increased anger word expression by younger adults to topic content rather than communication time progression. Although our research suggests topic influence outweighs time series effects, future studies should employ counterbalancing techniques to more precisely examine the impact of communication topics on emotional expression.

Research indicates that participant mental fatigue can diminish motivation as studies progress (Brosowsky et al., 2023). Moreover, anger is intrinsically linked to approach motivational systems (Carver & Harmon-Jones, 2009), suggesting that experimental duration may significantly impact emotional expression. Specifically, prolonged sessions could reduce participants' (particularly older adults') motivation during controversial issues stages, potentially dampening anger responses due to accumulated fatigue. Future research should incorporate adequate rest periods between experimental sessions to minimize fatigue-related biases in emotional expression.

The final limitation is that in Session A, both parties were required to discuss the story of an athlete. However, the familiarity and affinity toward athletes may vary between the interlocutors. Compared to older adults, young adults may resonate more with athletes and find it easier to discuss future-related topics (e.g., studying, graduating, employment), thereby amplifying the age differences in the use of future-focused words. Future research should investigate non-age-specific topics to verify the persistence of observed linguistic variations across diverse contextual frameworks.

Conclusions

This study invited both younger and older adults to engage in conversations on diverse topics and analyzed the content of their dialogues using the LIWC tool. The findings revealed that, in an intragenerational context, older adults expressed more present-focused words, fewer future-focused words, reduced negative emotional expressions, fewer cognitive process words, and more

family-related words compared to younger adults. These results indicate the presence of age-related differences in communication contexts, consistent with SST. Furthermore, intergenerational communication contexts facilitated linguistic consistency between younger and older adults in their expression of future-focused, cognitive process, and family-related words. These results demonstrate that verbal expression consistency across age groups plays a crucial role in intergenerational communication. Such findings can inform the design of future communication strategies aimed at enhancing communicative satisfaction across different age demographics.

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